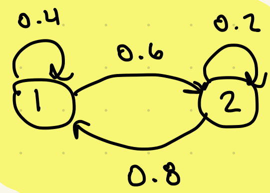
Assignment 7 Solutions

8.1



regular

A^1 has positive entries

$$A - I = \begin{bmatrix} -0.6 & 0.8 \\ 0.6 & -0.8 \end{bmatrix} \sim \begin{bmatrix} -3 & 4 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 \\ 0 & 0 \end{bmatrix}$$

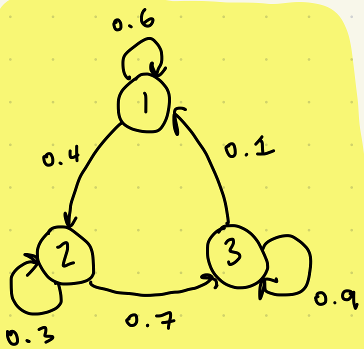
$x_1 = \frac{4}{3} x_2$

x_2 is free

$$x_1 + x_2 = 1 \Rightarrow \tfrac{4}{3} x_2 + x_2 = \tfrac{7}{3} x_2 = 1$$

SSV: $\begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix}$ unique

8.2



$$A^2 = \begin{bmatrix} 0.36 & 0.07 & 0.15 \\ 0.36 & 0.09 & 0.04 \\ 0.28 & 0.84 & 0.81 \end{bmatrix}$$

A^2 has positive entries

regular

$$A - I = \begin{bmatrix} -0.4 & 0 & 0.1 \\ 0.4 & -0.7 & 0 \\ 0 & 0.7 & -0.1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ -4^{+4} & 7 & 0^{-1} \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ 0 & 7 & -1 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ 0 & 7 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & -1/7 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = \frac{1}{4} x_3$

$x_2 = \frac{1}{7} x_3$

x_3 is free

$$x_1 + x_2 + x_3 = 1$$

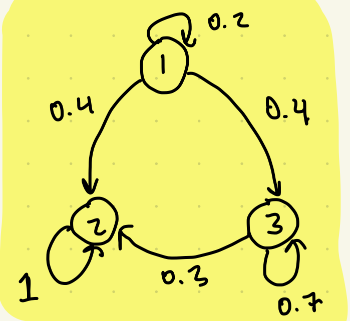
$$\tfrac{1}{4} x_3 + \tfrac{1}{7} x_3 + x_3 = 1$$

$$\frac{7 + 4 + 28}{28} x_3 = 1$$

$$\frac{39}{28} x_3 = 1 \qquad x_3 = \frac{28}{39}$$

SSV: $\begin{bmatrix} 7/39 \\ 4/39 \\ 28/39 \end{bmatrix}$ unique

8.5



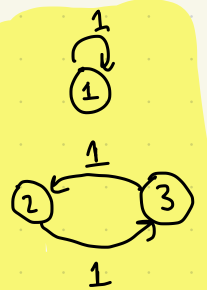
Not regular. $\underbrace{A^k \vec{e}_2}_{\text{second row of } A^k} = \vec{e}_2$

$$A - I = \begin{bmatrix} -0.8 & 0 & 0 \\ 0.4 & 0 & 0.3 \\ 0.4 & 0 & -0.3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 0$
$x_2 = 0$
x_3 is free

SSV: $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ unique

8.6



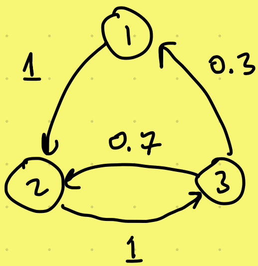
Not Regular. $A^k \vec{e}_1 = \vec{e}_1$

$$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free
$x_2 = x_3$
x_3 is free

SSV: $\begin{bmatrix} 0.4 \\ 0.3 \\ 0.3 \end{bmatrix}$ not unique, e.g., $\begin{bmatrix} 0.2 \\ 0.4 \\ 0.4 \end{bmatrix}$

8.8



$$A^5 = \begin{bmatrix} 0.21 & 0.09 & 0.147 \\ 0.49 & 0.42 & 0.433 \\ 0.3 & 0.49 & 0.42 \end{bmatrix}$$

has positive entries,

regular

$$A - I = \begin{bmatrix} -1 & 0 & 0.3 \\ 1^{-1} & -1 & 0.7^{+0.3} \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -0.3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 3/10\, x_3$

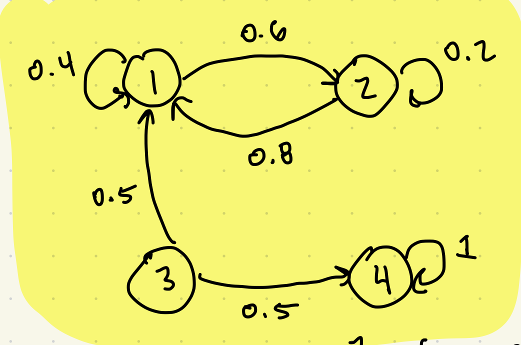
$x_2 = x_3$

x_3 is free

$$x_1 + x_2 + x_3 = 1 \qquad \frac{3}{10}x_3 + x_3 + x_3 = 1 \qquad \frac{23}{10}x_3 = 1$$

SSV: $\begin{bmatrix} 3/23 \\ 10/23 \\ 10/23 \end{bmatrix}$ unique

8.14



Not regular, $A^k \vec{e}_4 = \vec{e}_4$

$$A - I = \begin{bmatrix} -0.6 & 0.8 & 0.5 & 0 \\ 0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0.5 & 0 \end{bmatrix} \sim \begin{bmatrix} -6 & 8 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 4/3\, x_2$

x_2 is free

$x_3 = 0$

x_4 is free

$x_1 + x_2 + x_3 + x_4 = 1$

$4/3\, x_2 + x_2 = 1$

$\frac{7}{3} x_2 = 1$

SSV: $\begin{bmatrix} 4/7 \\ 3/7 \\ 0 \\ 0 \end{bmatrix}$ not unique

8.17 False $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

SSV: $\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$

8.20 True

$$AB = A[\vec{b}_1 \cdots \vec{b}_k]$$

$$= [A\vec{b}_1 \cdots A\vec{b}_k]$$

$$1^T(A\vec{b}_1) = 1^T \vec{b}_1 = 1$$

8.21 False $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

8.22 False (see 8.5)

8.24

$$\begin{array}{c|ccc} & L & S & H \\ \hline L & 0.6 & 0.1 & 0.05 \\ S & 0.3 & 0.7 & 0.4 \\ H & 0.1 & 0.2 & 0.55 \end{array}$$

$$A - I \sim \begin{bmatrix} 1 & 0 & -11/18 \\ 0 & 1 & -35/18 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 11/18\, x_3$

$x_2 = 35/18\, x_3$

x_3 is free

$x_1 + x_2 + x_3 = 1$ $\quad \frac{64}{18} x_3 = 1$

$x_3 = \frac{18}{64}$

$x_1 = 11/64$

$x_2 = 35/64$

$$\vec{s} = \begin{bmatrix} 11/64 \\ 35/64 \\ 18/64 \end{bmatrix}$$

Since transition matrix is regular,

$\lim_{k \to \infty} T^k \vec{e}_2 = \vec{s}$, which indicates it's most likely to stay steady in the long term

8.27

a) $\begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} = T$ (note: T is regular)

b) $T - I = \begin{bmatrix} p-1 & 1-p \\ 1-p & p-1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$

$x_1 = x_2$

x_2 is free

$$\lim_{k \to \infty} T^k \vec{e}_1 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$